Segment 04: Induction Motor(IM)
3-phase IM Construction, working principle, figure of equivalent circuit
Why is 1-phase IM not self-starting and explain its working with starting mechanism
Math: 34.3, 34.4, 34.5 (last 3)
Working of VFD with diagram/ How VFD controls the speed of IM.

Segment 05: Synchronous and Stepper Motor
Syn. Motor working principle, self-starting methods, speed control of synchronous motors using rectifiers and inverters
Define stepper motor, step angle calculation (math), full-step, 2-phase, half-step operation of variable reluctance stepper motor.

Segment 06: Measurement
static characteristics of measurement, 3-stage measurement procedure, all formulae (including error combinations) and maths

Segment 07:
Instrumentation amplifier figure, derivation, advantages(points), maths, temperature controller
Digital voltmeter, Digital frequency meter, Recorder

Segment 08:
Define Transducer, RTD, Thermocouple, Thermistor, Position Transducer all with relevant maths.
PV cell (working, materials and selection), Piezoelectric Transducer, Piezoelectric Ultrasonic Transducer